The Open Source Audit

A Capstone Project for OSS NGMC Course
Course: Open Source Software | Unit Coverage: 1 – 5

Software Audited: MySQL

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1. Introduction

This project is a structured audit of MySQL, one of the most widely deployed relational database management systems in the world. MySQL sits at the heart of the internet as we know it — it is the "M" in the famous LAMP stack (Linux, Apache, MySQL, PHP/Python/Perl) and powers millions of applications ranging from small personal blogs to massive enterprise platforms like Facebook, Twitter, and YouTube. Yet despite its ubiquity, many developers use MySQL without ever pausing to ask: who built this, why did they share it freely, and what freedoms does that sharing actually grant?

This audit answers those questions. It examines MySQL through five lenses: its origin story and the problem it was created to solve; the legal and philosophical implications of its license; the ethical questions surrounding open-source software; its practical footprint on a Linux system; and the broader ecosystem of tools and communities that surround it. Alongside the written analysis, five shell scripts demonstrate practical Linux skills and reinforce the connection between open-source philosophy and command-line practice.

The software chosen is **MySQL Community Edition** — the open-source version governed by the GNU General Public License version 2 (GPL v2). This choice is particularly rich because MySQL also exists as a commercial product, making it one of the most instructive examples of the dual-licensing model in all of open source.

Part A — Origin and Philosophy

A1. The Problem MySQL Was Created to Solve

To understand MySQL, one must travel back to the early 1990s, when the web was young and the commercial database market was controlled by a handful of expensive, proprietary giants. Oracle, Sybase, and Microsoft SQL Server dominated enterprise computing. These systems cost thousands of dollars in licensing fees, required specialised hardware, and were completely closed — you could not see how they worked, could not modify them, and were entirely dependent on the vendor for support. For small businesses, startups, and individual developers, a production-grade relational database was simply unaffordable.

The creators of MySQL — David Axmark, Allan Larsson, and Michael "Monty" Widenius — were Swedish programmers who had been building database tools professionally since the late 1980s. Their company, TcX DataKonsult AB, needed a fast, reliable database to power their own internal reporting and client applications. The existing free options were too slow or too limited. The commercial options were too expensive. So they built their own.

The first internal version of MySQL was written by Monty Widenius in 1994. It was not designed as an academic exercise or a philosophical statement — it was an intensely practical tool built out of frustration and necessity. The name "MySQL" combines "My," the name of Monty's daughter, with "SQL," the Structured Query Language that had become the universal standard for relational databases. In 1995, the team made a decision that would change the internet: they released MySQL publicly.

The timing was extraordinary. The World Wide Web was exploding. Web developers needed a way to store and retrieve data — user profiles, product listings, forum posts, comments. Apache HTTP Server had already made web serving free; PHP was making dynamic web pages accessible to anyone; Linux was providing a free operating system. MySQL became the final piece of what would become the LAMP stack. Suddenly, a teenager in a bedroom or a startup in a garage could build a database-backed web application without spending a single rupee on software licenses.

The specific technical problem MySQL was optimised to solve was read-heavy web workloads — applications that needed to retrieve data extremely fast, even at the cost of some transactional complexity. Early MySQL deliberately left out certain SQL features (like subqueries and stored procedures) to maintain speed. Critics called this a flaw; Monty called it a deliberate choice for its target audience. This pragmatic, performance-first philosophy is baked into MySQL's DNA to this day.

There is also a deeper story about trust. Proprietary databases were black boxes. When something went wrong, you filed a support ticket and waited. You could not audit the code for security vulnerabilities. You could not optimise the internals for your specific workload. By releasing the source code of MySQL, the founders gave the world something profound: the ability to understand, inspect, and improve the database managing their most valuable asset — their data. This transparency is not incidental to MySQL's success; it is central to it.

MySQL was acquired by Sun Microsystems in 2008 for approximately one billion US dollars — a figure that dramatically illustrated what a freely shared piece of software could be worth. When Oracle acquired Sun in 2010, concerns about MySQL's open-source future were so significant that the European Commission delayed approval of the acquisition to investigate. Monty Widenius himself lobbied against it and subsequently forked MySQL into MariaDB, a fully open-source continuation. This chain of events — a free tool becoming worth a billion dollars, and a community so invested in its freedom that they forked it to preserve that freedom — is perhaps the most compelling origin story in all of open source.

A2. The License — What It Actually Says

MySQL Community Edition is distributed under the **GNU General Public License version 2 (GPL v2)**. This is one of the most important and widely used open-source licenses in existence, authored by Richard Stallman as part of the GNU Project. Understanding what it actually says — not what people assume it says — is essential to understanding MySQL.

The Four Freedoms of Free Software

The Free Software Foundation defines free software through four essential freedoms:

- **Freedom 0:** The freedom to run the program for any purpose.
- **Freedom 1:** The freedom to study how the program works and adapt it to your needs. Access to source code is a precondition.
- **Freedom 2:** The freedom to redistribute copies so you can help your neighbour.
- **Freedom 3:** The freedom to distribute copies of your modified versions to others.

MySQL Community Edition grants all four freedoms. You can run it on any server, study its source code on GitHub, redistribute it to others, and publish your own modified version. This is genuine freedom — not a marketing term.

Can a Company Take MySQL, Modify It, and Sell It Without Sharing Changes?

This is where the GPL's most powerful and controversial clause comes into play: the **copyleft** provision. Under GPL v2, if you distribute a modified version of MySQL — whether for free or for payment — you must make the complete source code of your modifications available under the same GPL v2 license. You cannot take the code, add proprietary features, and sell the result as closed software while keeping your changes secret.

However, there is a critical distinction: if a company uses MySQL *internally* — for their own databases, their own applications — and never distributes the modified binary to anyone outside the organisation, the GPL does not require them to share anything. The obligation to share source code is triggered only by *distribution*. This is why companies like Google and Facebook can run heavily modified versions of MySQL internally without being legally obligated to release their changes.

For companies that want to distribute MySQL-based software without the copyleft obligation — for example, embedding MySQL in a commercial product — MySQL AB (and later Oracle) offers a commercial license. This is the famous **dual-licensing model**: the same software

under two different licenses, each serving a different market. The open-source community gets the GPL version for free; commercial vendors pay for the proprietary license. MySQL was one of the earliest and most successful examples of this model.

GPL vs MIT: A Philosophical Divide

The GPL and the MIT License represent two fundamentally different philosophies about software freedom:

Dimension	GPL v2	MIT License
Core philosophy	Copyleft — freedom must propagate	Permissive — freedom without obligation
Can you use in proprietary software?	Only if you do not distribute	Yes, without restriction
Must you share modifications?	Yes, if you distribute	No
Commercial use	Allowed (with conditions)	Allowed (without conditions)
Used by	Linux, Git, MySQL, WordPress	jQuery, React, .NET, Rails

If I were building something I intended to share with the world and wanted other developers to freely use and build upon — including in commercial products — I would choose the MIT License for its simplicity and permissiveness. However, if I were building a foundational tool and wanted to ensure that any improvements made by corporations were returned to the community, the GPL would be the right choice. The GPL is not just a license; it is a political statement that software freedom should be self-sustaining.

Free as in Free Beer vs Free as in Freedom

This distinction, popularised by Richard Stallman, is central to understanding MySQL. "Free as in free beer" (gratis) means you pay nothing. "Free as in freedom" (libre) means you have rights over the software — to use, study, modify, and redistribute it.

MySQL demonstrates both concepts, but they are not the same thing. MySQL Community Edition is free as in beer — you can download and use it without paying. But it is also free as in freedom — you have all four freedoms under the GPL. The commercial version of MySQL is free as in freedom only partially (source is available) but emphatically not free as in beer (you pay for it). The distinction matters enormously: a piece of software can be free to download but completely locked down (many "freeware" products), or it can cost money but give you full rights (some commercial GPL products).

A3. The Ethics of Open Source

Should All Software Be Open Source?

The case for: Knowledge has historically advanced fastest when it is shared openly. Open-source software creates a commons — a shared resource that anyone can draw from and contribute to. Security improves when code can be audited by thousands of independent eyes. Innovation accelerates when developers build on each other's work instead of reinventing foundations. There is also a democratic argument: software governs increasing amounts of public life — elections, medical records, financial systems. Society arguably has a right to inspect and understand the code running these systems.

The case against: Proprietary software funds the research and development that produces breakthrough innovations. Companies like Apple and Adobe have created transformative tools under closed licenses. Open source relies on the unpaid or underpaid labour of volunteers, which is not a sustainable model for every kind of software. Some software genuinely needs to remain secret — national security systems, medical devices with liability concerns, or tools whose security depends partly on obscurity. The world likely benefits from a mixed ecosystem rather than a mandate in either direction.

Is It Ethical for Companies to Profit from Free Community Labour?

This is one of the most contested questions in the technology world today. Consider the concrete example of Amazon Web Services (AWS), which runs RDS (Relational Database Service) built on MySQL and MariaDB, generating billions of dollars in revenue annually. For years, AWS contributed relatively little back to MySQL development. The community built the tool; Amazon packaged and sold access to it.

The GPL was specifically designed to prevent the most egregious forms of this: companies cannot take GPL code, add proprietary features, and distribute it without sharing changes. But the cloud-hosting model found a loophole — running GPL software as a service does not constitute "distribution," so the copyleft obligation never triggers. This led to the creation of new licenses like the AGPL (Affero GPL) and the SSPL (Server Side Public License), which close this loophole.

My view is that profiting from open-source software is ethically acceptable, but that companies which do so at scale have a moral — if not always a legal — obligation to contribute back. The healthiest open-source ecosystems, like Linux, are ones where major corporate beneficiaries (IBM, Google, Red Hat, Intel) are also major contributors.

Standing on the Shoulders of Giants

Isaac Newton's famous phrase resonates powerfully in software. Every application built today stands on layers of open-source foundations — the Linux kernel, the GCC compiler, the OpenSSL encryption library, the Python runtime, and yes, MySQL. These are not minor conveniences; they are the structural pillars of the digital world.

Open source enables this kind of accumulative innovation in a way that proprietary software cannot. When one company owns a foundation layer, every developer who builds on top of it is dependent on that company's continued goodwill, pricing decisions, and survival. Open-source foundations are owned by no one and available to everyone. This does not undermine original innovation — MySQL itself was an original innovation, and it enabled thousands of other original innovations. What it undermines is the artificial scarcity that proprietary licensing imposes on knowledge.

Part B — Linux Footprint

Installation Method

MySQL is available through the native package managers of all major Linux distributions, making installation straightforward without requiring compilation from source.

Ubuntu/Debian (apt):

```
sudo apt update
sudo apt install mysql-server
sudo systemctl start mysql
sudo systemctl enable mysql
```

RHEL/CentOS/Fedora (rpm/dnf):

```
sudo dnf install mysql-server # Fedora / RHEL 8+
sudo yum install mysql-server # CentOS 7
sudo systemctl start mysqld
sudo systemctl enable mysqld
```

Alternatively, MySQL provides official RPM and DEB packages through its own repository (repo.mysql.com), which always provides the latest stable version rather than the distribution-maintained version, which may lag behind.

Key Directories

Path	Purpose
`/usr/sbin/mysqld`	The MySQL server daemon binary
`/usr/bin/mysql`	The MySQL command-line client
`/etc/mysql/ or /etc/my.cnf`	Main configuration file(s)
`/var/lib/mysql/`	Data directory — where all databases are stored on disk
`/var/log/mysql/`	Log files (error log, general query log, slow query log)
`/run/mysqld/mysqld.sock`	Unix socket for local connections
`/usr/share/mysql/`	Shared resources, character sets, default config templates

User and Group

MySQL runs under a dedicated system user called **mysql**, which belongs to the **mysql** group. This is a standard security practice in Linux known as the *principle of least privilege*.

```
# Check the mysql user
id mysql
# Output: uid=999(mysql) gid=999(mysql) groups=999(mysql)

# The data directory is owned by this user
ls -la /var/lib/mysql
```

The mysql user has no login shell (typically /bin/false or /usr/sbin/nologin), which means even if an attacker compromised the MySQL process, they could not use that access to log into the

system interactively. The data directory /var/lib/mysql is owned by the mysql user and typically has permissions 750 or 700, preventing other users from reading the raw database files. This matters enormously for security in multi-user environments and shared hosting.

Service Management

MySQL integrates with systemd on modern Linux distributions, enabling standard service management commands:

```
sudo systemctl start mysql # Start the service
sudo systemctl stop mysql # Stop the service
sudo systemctl restart mysql # Restart (applies config changes)
sudo systemctl reload mysql # Reload config without full restart
sudo systemctl status mysql # Check if running + recent logs
sudo systemctl enable mysql # Auto-start on boot
sudo systemctl disable mysql # Remove auto-start
```

Note: On some distributions (particularly RHEL/CentOS), the service name is **mysqld** rather than **mysql**. The journalctl command can be used to inspect detailed logs: `sudo journalctl -u mysql --since today`

Update Model

Security updates to MySQL are delivered through the same package manager used for installation. On Ubuntu/Debian: `sudo apt upgrade mysql-server` will apply the latest security patches. The MySQL development team maintains multiple release branches simultaneously, with security patches back-ported to older stable versions. Critical vulnerabilities are assigned CVE (Common Vulnerabilities and Exposures) numbers and disclosed publicly — a transparency benefit that proprietary databases cannot match, since the community can verify that a fix is genuine and complete.

Part C — The FOSS Ecosystem

Dependencies

MySQL does not exist in isolation. It depends on a rich ecosystem of open-source libraries and tools:

- **glibc (GNU C Library):** The foundational C runtime that MySQL is compiled against on Linux. Everything from memory allocation to file I/O goes through glibc.
- **OpenSSL / LibreSSL:** Provides TLS/SSL encryption for client-server connections, ensuring data in transit is protected.
- **zlib:** Provides compression support used in MySQL's binary log compression and backup features.
- **libaio / io_uring:** Asynchronous I/O libraries that MySQL's InnoDB storage engine uses for high-performance disk operations.
- **Perl and Python:** Several MySQL utility scripts (like mysqldumpslow) are written in Perl; the MySQL connector for Python (mysql-connector-python) is a first-party library.
- **CMake:** The build system used to compile MySQL from source.

What MySQL Has Enabled and Inspired

MySQL's influence on the software ecosystem is difficult to overstate. Among its most significant offshoots and inspired projects:

- **MariaDB:** A direct fork of MySQL by the original creator, Monty Widenius, created after the Oracle acquisition. It is the default MySQL-compatible database in most Linux distributions today and is fully open-source under the GPL.
- **Percona Server for MySQL:** A drop-in replacement with enhanced performance and monitoring features, widely used in high-traffic production environments.
- **WordPress, Drupal, Joomla:** The three most widely used content management systems all use MySQL as their default database, collectively powering over 40% of all websites.
- **phpMyAdmin:** An open-source web-based GUI for MySQL administration, making the database accessible to users without command-line expertise.
- **Django and Laravel ORM:** Both of these popular web frameworks provide first-class support for MySQL, enabling rapid development of database-backed applications.
- **Vitess:** A MySQL-compatible database clustering solution, originally built at YouTube to scale MySQL to internet-scale workloads, now a CNCF graduated project.

MySQL in the LAMP Stack

The LAMP stack — Linux, Apache, MySQL, PHP — became the dominant architecture for dynamic web applications from the late 1990s through the 2010s. MySQL occupies the persistence layer: Apache handles HTTP requests, PHP generates dynamic content, and MySQL stores and retrieves the data that makes the content dynamic. The entire stack is free and open-source, which is precisely why it democratised web development. A teenager in

Bhopal in 2005 had access to the exact same database infrastructure as a Fortune 500 company — for free.

Community and Governance

MySQL's community structure is somewhat unusual because it is owned by Oracle, a commercial corporation, rather than a foundation. This creates tension. Oracle controls the roadmap and the commercial licensing, while the open-source community participates through:

- **MySQL Bugs Database (bugs.mysql.com):** Public bug tracking and reporting.
- **MySQL Community Forums (forums.mysql.com):** Official community support forums.
- **Stack Overflow:** The mysql tag has over 650,000 questions, making it one of the most actively answered technology communities in the world.
- **MySQL Conference & Expo (now Percona Live):** Annual technical conference bringing together MySQL users, contributors, and ecosystem vendors.
- **GitHub (github.com/mysql/mysql-server):** The official source code repository, where the community can read the code, file issues, and in limited cases submit patches.

Part D — Open Source vs Proprietary Comparison

The closest proprietary alternative to MySQL is **Microsoft SQL Server**. The comparison below is honest — open source is not always superior in every dimension.

Dimension	MySQL (Community)	Microsoft SQL Server
Cost	Free to download, use, and deploy. No per-seat	Developer Edition is free for development. Standard and Enterprise editions co
Security (code auditability)	Source code is fully public. Security researchers	Closed source. Security depends on Microsoft's internal audit teams and vulnerability disclosure
Support & reliability	Community support is vast (Stack Overflow, forums). Oracle offers paid support for enterprise customers	Microsoft offers paid support plans. Integration with Azure, Active Directory, and Power BI
Freedom to modify	Full source access under GPL. Can modify, rec	No access to source code. Customers must rely on Microsoft for bug fixes and feature requests
Community vs corporate control	Owned by Oracle (commercial), but with an acti	Fully controlled by Microsoft. No community-governed fork. Microsoft's commercial interests drive the roadmap
Overall verdict	Best for web applications, open environments,	Best for Windows-centric enterprises with heavy Microsoft stack investment

Verdict and Recommendation

Based on this audit, I would recommend MySQL Community Edition for the vast majority of real-world web application deployments, particularly for organisations that value transparency, cost efficiency, and long-term independence from vendor lock-in. MySQL's combination of proven reliability (it powers companies like Facebook, GitHub, and Airbnb at astronomical scale), active community support, and zero licensing cost makes it the rational default choice for most applications. The existence of MariaDB as a fully community-controlled fork provides an important insurance policy against Oracle's commercial interests ever diverging too far from community needs.

The one domain where I would hesitate to recommend MySQL over a proprietary alternative is highly complex analytical workloads in a Microsoft-heavy enterprise environment, where SQL Server's integration with Power BI, Azure, and Active Directory may justify its cost. As for contributing: yes, I would contribute to the MariaDB fork rather than MySQL itself, as the fully community-governed nature of MariaDB better aligns with the open-source philosophy this course has explored. Contributing back to the tools we depend on is not just good citizenship — it is how the open-source commons remains healthy for the next generation of developers.

Shell Scripts — Overview and Documentation

Five shell scripts accompany this report, each targeting specific Linux and shell scripting concepts. Every script is submitted as a plain .sh file and is fully commented. Below, each script is documented with its code, a description of its purpose, and the shell concepts it demonstrates.

Script 1 — System Identity Report

Shell Concepts Demonstrated: Variables, command substitution $(), echo, output formatting, reading /etc/os-release

Description: This script acts as a welcome screen for the Linux system. It collects and displays the Linux distribution name and kernel version, the current logged-in user and their home directory, system uptime, the current date and time, and a message about the open-source license covering the OS (GPL v2 for the Linux kernel). It uses command substitution to capture the output of system commands like uname, whoami, uptime, and date into variables, then formats them for clean display using echo.

Script Code:

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Samarth Verma| Reg: 24BCE10139
# Course: Open Source Software (OSS NGMC)
# Purpose: Display a formatted system identity welcome screen

# ■■■ Variables ■■■
```

```
# Read distro name from the standard os-release file
if [ -f /etc/os-release ]; then
    # Source the file to load its variables
    source /etc/os-release
    DISTRO_NAME="$PRETTY_NAME"
else
    # Fallback if file is missing
    DISTRO_NAME="Unknown Linux Distribution"
fi

# ■■■ Display ■■■
```

Description: This script checks whether MySQL (or any specified package) is installed on the system, prints its version and license information, and uses a case statement to display a short philosophical description of the software. It auto-detects whether the system uses apt (Debian/Ubuntu) or rpm (RHEL/Fedora) and runs the appropriate command. The case statement covers MySQL and several other common FOSS packages.

Script Code:

```
#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: Samarth Verma | Reg: 24BCE10139
# Course: Open Source Software (OSS NGMC)
# Purpose: Check if a FOSS package is installed, show its info,
#          and display a philosophy note about the software.

# ■■■ Configuration ■■■
```

```
# ■■■ Check installation and display info ■■■■■■■■■■■■■■■■■■■■■■■■
if [ "$PKG_MANAGER" = "rpm" ]; then
    # RPM-based system (RHEL, Fedora, CentOS)
    if rpm -q "$PACKAGE" &>/dev/null; then
        echo "  STATUS: $PACKAGE IS INSTALLED"
        echo ""
        echo "  Package Details:"
        # Pipe rpm output through grep to show only key fields
        rpm -qi "$PACKAGE" | grep -E "^(Version|License|Summary|URL)" |
     while IFS= read -r line; do echo "    $line"; done
    else
        echo "  STATUS: $PACKAGE is NOT installed."
        echo "  Install with: sudo dnf install $PACKAGE"
    fi

elif [ "$PKG_MANAGER" = "dpkg" ]; then
    # APT-based system (Ubuntu, Debian)
    if dpkg -l "$PACKAGE" 2>/dev/null | grep -q "^ii"; then
        echo "  STATUS: $PACKAGE IS INSTALLED"
        echo ""
        # Show version using dpkg-query
        VERSION=$(dpkg-query -W -f='${Version}' "$PACKAGE" 2>/dev/null)
        echo "    Version : $VERSION"
        echo "    License : GPL v2 (GNU General Public License version 2)"
        echo "    Summary : MySQL database server"
    else
        echo "  STATUS: $PACKAGE is NOT installed."
        echo "  Install with: sudo apt install $PACKAGE"
    fi
fi

echo ""
echo "==================================================="
echo "  FOSS Philosophy Notes"
echo "==================================================="

# ■■■ Case statement: philosophy notes ■■■■■■■■■■■■■■■■■■■■■■■■■■
```

```
# Match on common package names and print a philosophy note
case "$PACKAGE" in
    mysql-server|mysql|mysqld)
        echo "  MySQL: The database at the heart of the web — GPL v2"
        echo "  means the community keeps the right to study and share it."
        ;;
    httpd|apache2)
        echo "  Apache HTTP Server: Built the open internet. The Apache"
        echo "  License allows anyone to use it, even in proprietary products."
        ;;
    firefox)
        echo "  Firefox: A nonprofit browser fighting for an open, private web.
"
        echo "  Mozilla's mission proves open source can challenge corporate gi
ants."
        ;;
    vlc)
        echo "  VLC: Built by students in Paris who just wanted to stream video
."
        echo "  LGPL means it can be embedded in other open or closed software.
"
        ;;
    git)
        echo "  Git: Linus Torvalds built this when BitKeeper's free license wa
s"
        echo "  revoked. A reminder that open source protects us from lock-in."
        ;;
    python3|python)
        echo "  Python: Shaped entirely by community consensus. The PSF License
"
        echo "  is permissive — freedom without copyleft obligations."
        ;;
    libreoffice)
        echo "  LibreOffice: Born from a community fork of OpenOffice when Orac
le"
        echo "  acquired it. Proof that the community will fork to preserve fre
edom."
        ;;
    *)
```

```
        echo "  $PACKAGE: A free and open-source software package."
        echo "  Check its license at: https://spdx.org/licenses/"
        ;;
esac

echo ""
echo "  Script complete."
```

Script 3 — Disk and Permission Auditor

Shell Concepts Demonstrated: for loop, arrays, du, ls -ld, awk, cut, directory existence check, MySQL config path

Description: This script loops through a predefined list of important system directories and reports the permissions, owner, group, and disk usage of each. It uses a for loop over a bash array, the ls -ld command piped through awk to extract fields, and du -sh for human-readable disk usage. At the end, it specifically checks for MySQL's configuration and data directories and reports their permissions — directly relevant to MySQL's security model discussed in Part B.

Script Code:

```
#!/bin/bash
# Script 3: Disk and Permission Auditor
# Author: Samarth Verma | Reg: 24BCE10139
# Course: Open Source Software (OSS NGMC)
# Purpose: Audit permissions and disk usage of key system directories,
#          then specifically check MySQL-related directories.

# ■■■ Define directories to audit ■■■
```

```
echo "  ━━━
```

```
    "/usr/sbin/mysqld"
)

for PATH_ITEM in "${MYSQL_PATHS[@]}"; do
    # Check for both files and directories
    if [ -e "$PATH_ITEM" ]; then
        PERMS=$(ls -ld "$PATH_ITEM" | awk '{print $1}')
        OWNER=$(ls -ld "$PATH_ITEM" | awk '{print $3}')
        GROUP=$(ls -ld "$PATH_ITEM" | awk '{print $4}')
        echo "  FOUND   : $PATH_ITEM"
        echo "            Owner: $OWNER | Group: $GROUP | Permissions: $PERMS"

        # Security note for the data directory
        if [ "$PATH_ITEM" = "/var/lib/mysql" ]; then
            echo "            NOTE: This is MySQL's data directory. It should b
e"
            echo "            owned by the 'mysql' user for security."
        fi
        echo ""
    else
        echo "  MISSING : $PATH_ITEM (MySQL may not be installed)"
        echo ""
    fi
done

echo "  Audit complete."
```

Script 4 — Log File Analyzer

Shell Concepts Demonstrated: while read loop, if-then, counter variables, command-line arguments ($1, $2), tail, grep, exit codes

Description: This script reads a log file line by line using a while read loop, counts occurrences of a keyword (defaulting to "error" if not specified), and prints a summary. It accepts the log file path and optional keyword as command-line arguments. It validates that the file exists before processing. At the end, it prints the last 5 matching lines using grep and tail. A retry loop demonstrates do-while style logic for handling empty files.

Script Code:

```
#!/bin/bash
# Script 4: Log File Analyzer
# Author:Samarth Verma| Reg: 24BCE10139

# Course: Open Source Software (OSS NGMC)
# Purpose: Read a log file line-by-line, count keyword occurrences,
#          and display a summary with the last 5 matching lines.
# Usage: ./log_analyzer.sh /path/to/logfile [keyword]
                                                                    ■■■■■
# ■■■ Command-line arguments ■■■■■
```

```
    exit 1
fi

# ■■■ Retry logic for empty file ■■■
```

```
# ■■■ Summary output ■■■
```

Script Code:

```
#!/bin/bash
# Script 5: Open Source Manifesto Generator
# Author:Samarth Verma| Reg: 24BCE10139

# Course: Open Source Software (OSS NGMC)
# Purpose: Generate a personalised open-source philosophy statement
#          by collecting user input and composing a reflective paragraph.

# ■■■ Alias concept demonstration (comment) ■■■■■■■■■■■■■■■■■■■■■
# In a live shell session, you could define a shortcut alias:
#   alias mkmanifesto='bash ./manifesto_generator.sh'
# Aliases are defined in ~/.bashrc for persistence across sessions.
# They differ from scripts in that they exist only in the shell session.

echo "==================================================="
echo "   Open Source Manifesto Generator"
echo "   Answer three questions to generate your manifesto."
echo "==================================================="
echo ""
■
# ■■■ Collect user input with read -p ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
# The -p flag displays a prompt on the same line before reading input
read -p "  1. Name one open-source tool you use every day: " TOOL

read -p "  2. In one word, what does 'freedom' mean to you?  " FREEDOM

read -p "  3. Name one thing you would build and share freely: " BUILD

# ■■■ Validate that all inputs were provided ■■■■■■■■■■■■■■■■■■■■
if [ -z "$TOOL" ] || [ -z "$FREEDOM" ] || [ -z "$BUILD" ]; then
    echo ""
    echo "  ERROR: All three questions must be answered."
    exit 1
fi
■■■■■■■
# ■■■ Prepare metadata ■■■■■■■■■■■■■■■■■■■
```

```
# Capture the current date in a readable format using the date command
DATE=$(date '+%d %B %Y')
# The output filename includes the current username for personalisation
OUTPUT="manifesto_$(whoami).txt"

echo ""
echo "  Composing your manifesto..."
echo ""

# ■■■ Compose the manifesto paragraph ■■■
```

```
PARA3="$PARA3 who needs it, wherever they are."

PARA4="Open source is not just a licensing model. It is a statement about how"
PARA4="$PARA4 knowledge should work: accumulated, shared, and improved by every one."
PARA4="$PARA4 I stand on the shoulders of giants, and I intend to add to that foundation."

# Write paragraphs to the file, each separated by a blank line
echo "$PARA1" >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "$PARA2" >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "$PARA3" >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "$PARA4" >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "========================================" >> "$OUTPUT"

# ■■■ Display result ■■■
```

7. Conclusion

This audit of MySQL has been an exercise in looking past the surface of a familiar tool to understand what it actually represents. MySQL is not just a piece of software — it is a product of a specific moment in history when a group of pragmatic Swedish programmers decided that a tool they needed should be shared with the world. That decision, multiplied across thousands of similar decisions by thousands of developers, created the internet as we know it.

The GPL v2 license that governs MySQL is not merely a legal document — it is a philosophical statement about power and accountability in technology. Copyleft ensures that the freedom to use, study, and improve software cannot be taken away by commercial actors. The dual-licensing model that MySQL pioneered shows that this freedom and commercial sustainability are not mutually exclusive — they can coexist.

The Linux footprint analysis revealed how carefully MySQL integrates into the operating system — the dedicated mysql user, the clean separation of binaries, config, data, and logs, and the systemd integration all reflect decades of accumulated wisdom about how server software should behave on Linux. These are not accidental choices; they are the product of an open-source community that can inspect, debate, and improve every aspect of how the software works.

The ecosystem analysis showed that MySQL's most important contribution is not the code itself, but what it enabled. WordPress, Drupal, the LAMP stack, and hundreds of frameworks and tools exist because MySQL made the database layer free. The fork that produced MariaDB shows that open-source communities are not passive consumers — they are active custodians of the tools they depend on, willing to act when those tools are threatened.

Most importantly, this project has made concrete what was previously abstract. Open source is not a technical detail or a licensing footnote. It is the reason that a student in Bhopal today has access to the same database technology as an engineer at Google. That equality of access — that freedom — is worth understanding, worth protecting, and worth contributing to.

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